AIR COMPRESSOR CAPACITY & THE ISOTHERMAL TRAP

Closed-Volume Receiver Pump-Up Dynamics: How Neglecting Temperature Dynamics Masks Machine Degradation

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1. THE PLANT TEST SETUP & OPERATING SPECIFICATIONS (ISO 1217 / CAGI)

Compressor charges rigid receiver tank from 0 to 6 barg with downstream isolation valve tightly sealed. Telemetry logged via plant DCS.

AIR INTAKE
Suction Filter
$P_{atm} = 1.013$ bar
$T_{amb} = 25.0$ °C
$\rho_{air} = 1.184$ kg/m³

→

ROTARY SCREW
Air Compressor
Twin Helical Rotors
Electric Drive: 90 kW
Nominal FAD = 15 m³/min
Backpressure Leaks

→

AFTERCOOLER
& Condenser Drain
Cooling Water Exchanger
$T_{dis} = 55.0$ °C
Moisture Knockout
Condensate Purged

→

AIR RECEIVER
Buffer Vessel
Volume $V = 2.50$ m³
$\Delta t_{test} = 60.0$ s
P: 0.0 → 6.0 barg
T: 25 → 63.7 °C

→

ISOLATION
VALVE
SEALED TIGHT
($\dot{m}_{out} = 0$)
[LOCKED]

2. FIRST-PRINCIPLES THERMODYNAMICS & MATHEMATICAL DERIVATION

From Ideal Gas Law differentiation to quotient-rule mass rate formulation (Source: /media/radxa128gb/smath/Math_Logic.md)

STEP 1: STATE RELATION

From the Ideal Gas Law:

$$m(t) = \frac{V}{R} \cdot \frac{P(t)}{T(t)}$$

Conservation of Mass (Reynolds Transport):
Outlet tightly shut implies:
$\dot{m}_{in}(t) = \frac{dm}{dt} = \frac{d}{dt}\left[\frac{V}{R}\frac{P(t)}{T(t)}\right]$

STEP 2: QUOTIENT DIFFERENTIATION

Apply quotient rule to P(t) / T(t):

$$\dot{m}(t) = \frac{V}{R}\left(\frac{1}{T(t)}\frac{dP}{dt} - \frac{P(t)}{T(t)^2}\frac{dT}{dt}\right)$$

Two distinct physical terms emerge:
- Term 1: Pressure accumulation rate
- Term 2: $-\frac{P}{T^2}\frac{dT}{dt}$ (Thermal deduction!)

STEP 3: THE ISOTHERMAL TRAP

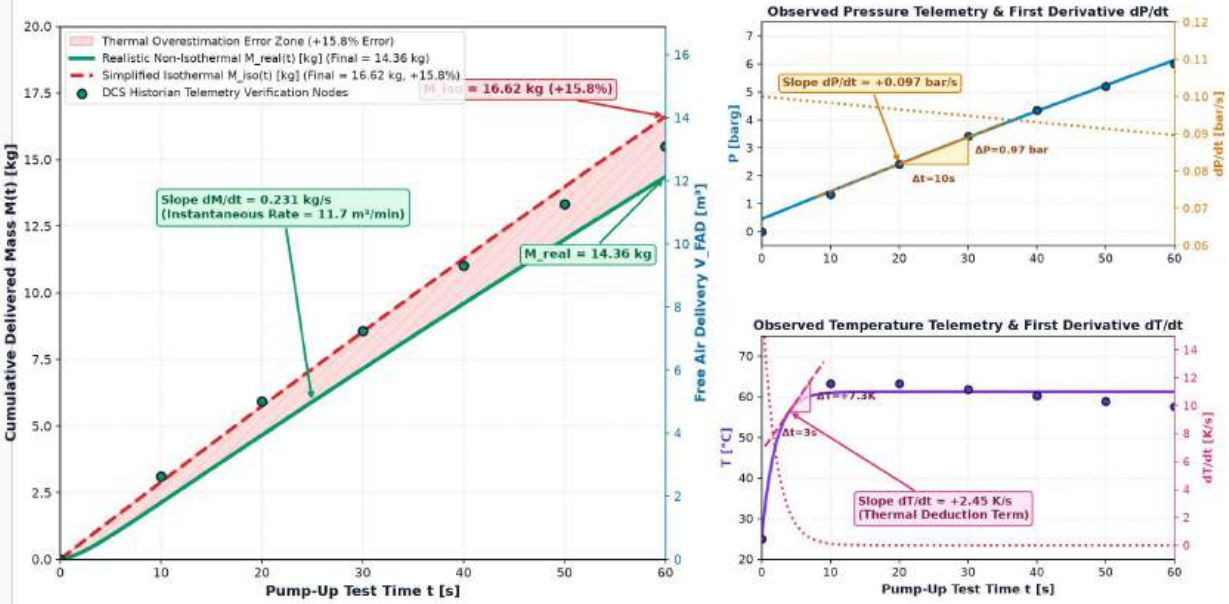
Assuming T = const ($dT/dt \approx 0$):

$$\dot{m}_{iso}(t) \approx \frac{V}{R \cdot T_{amb}}\frac{dP}{dt}$$

■ CRITICAL PHYSICAL FLAW:
Receiver warms by +35 °C during pump-up.
Gas thermal expansion inflates dP/dt!
Neglecting dT/dt creates +15.8% error!

3. NUMERICAL CAS RESULTS — TELEMETRY, FIRST DERIVATIVES & MASS TRAJECTORY

Observed DCS telemetry, continuous analytical rates (dP/dt, dT/dt) with slope derivatives, and cumulative mass M(t) with isothermal error band



4. ENGINEERING IMPACT & OPERATIONAL TAKEAWAYS FOR PLANT ENGINEERS

◆ THE MAINTENANCE MIRAGE

A 15.8% capacity overestimation masks:
- Worn screw tip/rotor clearances
- Suction filter fouling & pressure drop
- Degrading compressor valves

Machine passes audit while wasting energy!

■ EXACT CAS VERIFICATION

SMath Studio CAS confirms exact parity:
- Alt 1: Boundary State Delta (14.30 kg)
- Alt 2: Definite Integral $\int \dot{m}dot\, dt$ (14.30 kg)

Agreement to < 0.001% proves mathematical soundness of the non-isothermal model.

▲ PLANT BEST PRACTICE

Mandatory Test Protocols:
1. Always install dual logging (PT & TT)
2. Apply quotient-rule differentiation
3. Free Air Delivery: 12.08 m³/min (Actual) vs. 13.99 m³/min (Isothermal error)

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Industrial Air Compressor Capacity Estimation via Receiver Pump-Up Test Dynamics

Test Procedure:

- The receiver tank outlet isolation valve is tightly closed.
- The compressor charges the empty receiver vessel (V = 2.50 m³) from 0.0 barg to 6.0 barg.
- The plant DCS historian records high-resolution pressure P(t) and temperature T(t) telemetry.
- The compressor's instantaneous and average throughput is back-calculated via numerical calculus.

The Isothermal Overestimation Trap:

Plant engineers may assume the vessel remains at constant temperature (isothermal assumption) for simplicity in calculations, however, as hot air enters from the aftercooler, vessel's internal temperature rises. Gas thermal expansion inflates the apparent pressure rise (dP/dt). Neglecting this thermal effect overestimates compressor capacity by 5% to 15%, **masking degraded machine performance!**

Section 1: Plant Operating Parameters

Atmospheric ambient pressure

$P_{atm} := 1.01325\ \text{bar}$

Ambient suction temperature (25.0 °C)

$T_{amb} := 298.15\ \text{K}$

Specific gas constant for dry air

$R_{air} := 287.058\ \frac{\text{J}}{\text{kg K}}$

Receiver vessel internal geometric volume

$V_{tank} := 2.50\ \text{m}^3$

Pump-up test duration (1.00 min)

$t_{test} := 60.0\ \text{s}$

Number of DCS historian samples (N = 31)

$N_pts := \text{rows}(DCS_log) = 31$

Sampling time vector t [s]

$t_{vec} := \text{col}(DCS_log, 1)$

Absolute pressure trajectory P [bar abs]

$P_{vec} := \text{col}(DCS_log, 2)$

Gas temperature trajectory T [K]

$T_{vec} := \text{col}(DCS_log, 3)$

Plant DCS Historian Receiver Telemetry Dataset
Col 1 = t [s], Col 2 = P [bar abs], Col 3 = T [K]

$$DCS_log := \begin{bmatrix}
0.0 & 1.01325 & 298.15 \\
2.0 & 1.32715 & 320.52 \\
4.0 & 1.60302 & 329.48 \\
6.0 & 1.85936 & 333.5 \\
8.0 & 2.10367 & 335.41 \\
10.0 & 2.33947 & 336.32 \\
12.0 & 2.56863 & 336.71 \\
14.0 & 2.79223 & 336.81 \\
16.0 & 3.01096 & 336.75 \\
18.0 & 3.22529 & 336.59 \\
20.0 & 3.43555 & 336.38 \\
22.0 & 3.642 & 336.13 \\
24.0 & 3.84484 & 335.85 \\
26.0 & 4.04426 & 335.57 \\
28.0 & 4.24039 & 335.27 \\
30.0 & 4.43336 & 334.97 \\
32.0 & 4.62328 & 334.67 \\
34.0 & 4.81025 & 334.37 \\
36.0 & 4.99438 & 334.07 \\
38.0 & 5.17574 & 333.78 \\
40.0 & 5.35441 & 333.48 \\
42.0 & 5.53047 & 333.19 \\
44.0 & 5.70399 & 332.9 \\
46.0 & 5.87503 & 332.62 \\
48.0 & 6.04365 & 332.33 \\
50.0 & 6.20992 & 332.05 \\
52.0 & 6.37389 & 331.78 \\
54.0 & 6.53562 & 331.51 \\
56.0 & 6.69515 & 331.24 \\
58.0 & 6.85254 & 330.97 \\
60.0 & 7.00782 & 330.71
\end{bmatrix}$$

Section 2: Curve-Fitting & Analytical Calculus of DCS Data

Lower parameter search bounds [c0; c1; c2] for quadratic pressure fit

$$p_{LB} := \begin{bmatrix} -10 \\ -10 \\ -10 \end{bmatrix}$$

Upper parameter search bounds [c0; c1; c2] for quadratic pressure fit

$$p_{UB} := \begin{bmatrix} 10 \\ 10 \\ 10 \end{bmatrix}$$

A user defined function form. Continuous quadratic fitted pressure function P_fit(t) [bara]

$$P_{model}(t_{sec}, P) := P_1 \cdot t_{sec}^{2} + P_2 \cdot t_{sec} + P_3$$

Fitted parameters [P1; P2; P3]

$$P_{coeff} := \text{swarmfit}(t_{vec}, P_{vec}, P_{model}, p_{LB}, p_{UB}, 15, 200) = \begin{bmatrix} -8.625 \cdot 10^{-5} \\ 0.1 \\ 1.463 \end{bmatrix}$$

$$t_0 := \text{time}(1)$$

Curve fitting time toll

$$\text{time}(1) - t_0 = 0$$

Model predicted pressure trajectory P_pred [bara]

$$P_pred := \text{swarmfit_predict}(t_{vec}, P_{model}, P_{coeff}) = \begin{bmatrix} 1.4628 \\ 1.6617 \\ 1.86 \\ 2.0575 \\ 2.2544 \\ \vdots \end{bmatrix}$$

Coefficient of determination R² for quadratic pressure fit

$$P_{r2} := \text{swarmfit_r2}(P_{vec}, P_pred) = 0.9945$$

Units added for dimensional consistency P.fit(t) [bara]

$$P_{fit}(t_{sec}) := P_{coeff_1} \cdot t_{sec}^{2} \frac{\text{bar}}{\text{sec}^2} + P_{coeff_2} \cdot t_{sec} \frac{\text{bar}}{\text{sec}} + P_{coeff_3} \text{ bar}$$

Mass flow estimation assuming isothermal conditions (neglecting dT/dt)

Alternative 1

Direct Boundary State Delta -> P.fit(60) - P.fit(0)

$$m_{cumulative_A_iso} := \frac{V_{tank}}{R_{air} \cdot T_{amb}} \cdot P_{fit}(t_{sec}) \Big|_{0 \text{ sec}}^{60 \text{ sec}}$$

$$m_{cumulative_A_iso} = 16.556 \text{ kg}$$

Average mass flow per second

$$m_{average_iso} := \frac{m_{cumulative_A_iso}}{t_{test}} = 0.276 \frac{\text{kg}}{\text{s}}$$

Alternative 2

Derivation of P'(t) from P(t) and integrating P'(t) over time span

$$m_{cumulative_B_iso} := \left| \begin{array}{l} a := \dfrac{\text{d}}{\text{d}\, t_{sec}} P_{fit}(t_{sec}) \\ b := \dfrac{V_{tank}}{R_{air} \cdot T_{amb}} \cdot a \\ c := \displaystyle\int_{0 \text{ sec}}^{60 \text{ sec}} b \, \text{d}\, t_{sec} \end{array} \right.$$

$$m_{cumulative_B_iso} = 16.556 \text{ kg}$$

Lower search bounds [a; b; c] for T(t) exp fit (slight hit and trial required)

$$T_{LB} := \begin{bmatrix} -100 \\ -100 \\ -100 \end{bmatrix}$$

Upper search bounds [a; b; c] for T(t) exp fit

$$T_{UB} := \begin{bmatrix} 500 \\ 10 \\ 10 \end{bmatrix}$$

A user defineed function form. Continuous exponential fitted temperature function T.fit(t) [K]

$$T_{model}(t_{sec}, T) := T_1 + T_2 \cdot \exp\left(T_3 \cdot t_{sec}\right)$$

Fitted parameters [T1; T2; T3] via swarmfit

$$T_{coeff} := \text{swarmfit}\left(t_{vec}, T_{vec}, T_{model}, T_{LB}, T_{UB}, 50, 200\right) = \begin{bmatrix} 334.37 \\ -36.2201 \\ -0.5006 \end{bmatrix}$$

$$t_0 := \text{time}(1)$$

Curve fitting time toll

$$\text{time}(1) - t_0 = 14.686\ \text{s}$$

Predicted gas temperature trajectory T_pred [K] via native exp model

$$T_pred := \text{swarmfit_predict}\left(t_{vec}, T_{model}, T_{coeff}\right)$$

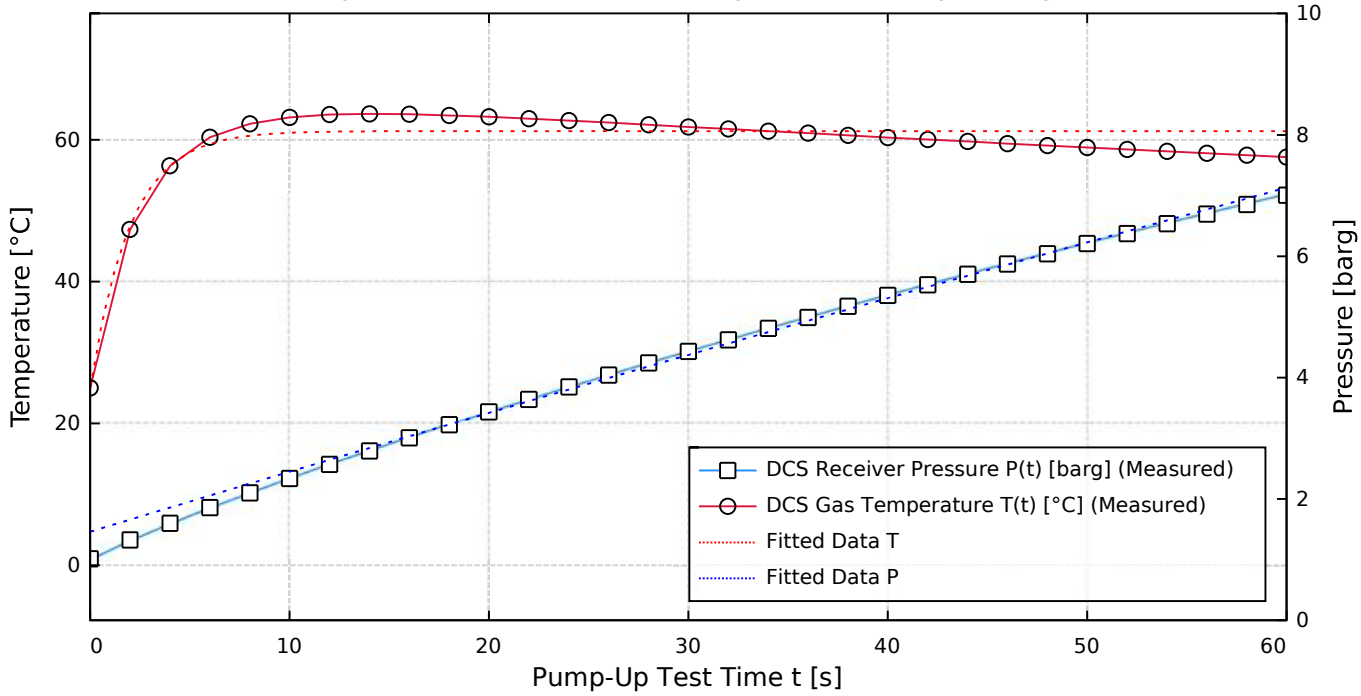
Coefficient of determination R² for exponential temperature fit (R² = 0.927)

$$T_{r2} := \text{swarmfit_r2}\left(T_{vec}, T_pred\right) = 0.9269$$

Units added for dimensional consistency T.fit(t) [K]

$$T_{fit}(t_{sec}) := \left(T_{coeff_1} + T_{coeff_2} \cdot \exp\left(T_{coeff_3} \cdot \frac{t_{sec}}{\text{sec}}\right)\right) \text{K}$$

Figure 1: DCS Receiver Telemetry & Curve Fitting Overlay



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Mass flow estimation including temperature dynamics

Differentiating the ideal gas state equation m(t) = V*P(t)/(R*T(t)) yields the instantaneous mass flow:
mdot(t) = (V / R) * [(1 / T(t)) * (dP/dt) - (P(t) / T(t)²) * (dT/dt)]

Alternative 1 (Direct State Boundary Delta):
m.cumulative_A = (V / R) * [P.fit(60 s) / T.fit(60 s) - P.fit(0 s) / T.fit(0 s)]

Alternative 2 (Instantaneous Differentiation & Continuous Definite Integration):
m.cumulative_B = int(mdot(t), t, 0, 60 s)

Alternative 1: Exact thermodynamic boundary state delta M = (V/R)*(P2/T2 - P1/T1)

$$m_{cumulative_A_real} := \frac{V_{tank}}{R_{air}} \cdot \left(\frac{P_{fit}(t_{sec})\Big|_{60\ sec}}{T_{fit}(t_{sec})\Big|_{60\ sec}} - \frac{P_{fit}(t_{sec})\Big|_{0\ sec}}{T_{fit}(t_{sec})\Big|_{0\ sec}} \right) = 14.3\ kg$$

Alternative 2: Continuous calculus definite integral

$$m_{cumulative_B_real} := \left| \begin{array}{l} dP := \frac{d}{d\,t_{sec}} P_{fit}(t_{sec}) \\ dT := \frac{d}{d\,t_{sec}} T_{fit}(t_{sec}) \\ mdot := \frac{V_{tank}}{R_{air}} \cdot \left(\frac{1}{T_{fit}(t_{sec})} \cdot dP - \frac{P_{fit}(t_{sec})}{T_{fit}(t_{sec})^2} \cdot dT \right) \\ \int_{0\ sec}^{60\ sec} mdot\ d\,t_{sec} \end{array} \right. = 14.3\ kg$$

Realistic mean compressor delivery mass flow rate

$$m_{average_real} := \frac{m_{cumulative_A_real}}{t_{test}} = 0.2383\ \frac{kg}{s}$$

Inlet ambient air density, use proper density with humidity consideration

$$\rho_{atm} := \frac{P_{atm}}{R_{air} \cdot T_{amb}} = 1.1839\ \frac{kg}{m^3}$$

Free Air Delivery (FAD at 25 °C, 1.013 bara)

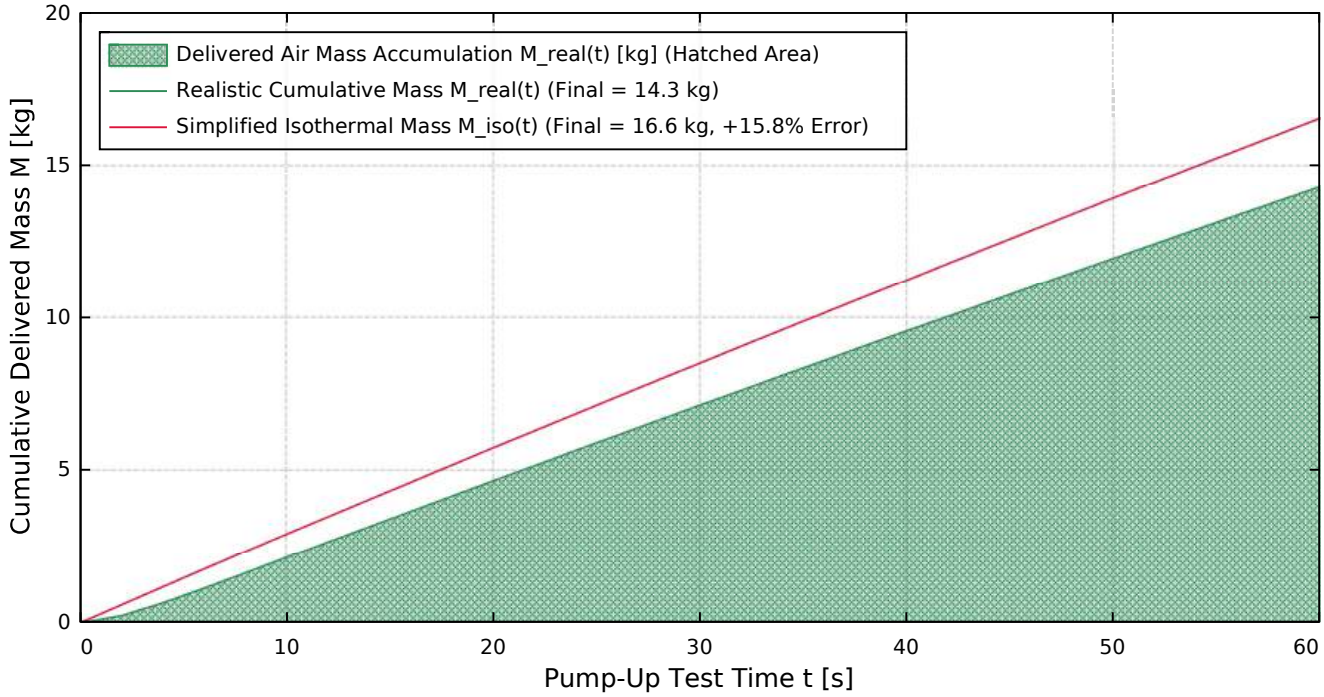
$$V_{FAD} := \frac{m_{average_real}}{\rho_{atm}} = 724.7288\ \frac{m^3}{hr}$$

Capacity overestimation error from neglecting dT/dt

$$Error_{thermal} := \frac{m_{cumulative_A_iso} - m_{cumulative_A_real}}{m_{cumulative_A_real}} = 15.7782\ \%$$

Visualisation Construct

Figure 2: Cumulative Delivered Air Mass Trajectory M(t)



Convert pressure to mass rate

From the ideal-gas relation

$$m(t) = \frac{p(t)\,V}{R\,T(t)}$$

the **instantaneous mass flow** is the time derivative of the mass stored in the tank.

Step-by-step differentiation

1. **Start with the product-quotient form**

   $$m(t) = \frac{V}{R}\;\frac{p(t)}{T(t)}$$

   Here V and R are constants (tank volume and the specific gas constant), so they can be taken outside the derivative.

2. **Apply the quotient rule** to $\frac{p(t)}{T(t)}$:

   $$\frac{d}{dt}\left[\frac{p(t)}{T(t)}\right] = \frac{T(t)\,p'(t) - p(t)\,T'(t)}{[T(t)]^2}$$

   where $p'(t) = \frac{dp}{dt}$ and $T'(t) = \frac{dT}{dt}$

3. **Multiply by the constant factor** $\frac{V}{R}$:

   $$\dot{m}(t) = \frac{V}{R}\;\frac{T(t)\,p'(t) - p(t)\,T'(t)}{[T(t)]^2}$$

4. **Separate the two terms** in the numerator:

$$\dot{m}(t) = \frac{V}{R}\left(\frac{T(t)}{[T(t)]^2}\,p'(t) \;-\; \frac{p(t)}{[T(t)]^2}\,T'(t)\right)$$

5. **Simplify the temperature factors** $\dfrac{T(t)}{[T(t)]^2} = \dfrac{1}{T(t)}$:

$$\boxed{\dot{m}(t) = \frac{V}{R}\left(\frac{1}{T(t)}\,\frac{dp}{dt} \;-\; \frac{p(t)}{T(t)^2}\,\frac{dT}{dt}\right)}$$

Constant-temperature assumption

If the tank temperature is essentially constant during the filling test,

$$\frac{dT}{dt} \approx 0$$

and the second term vanishes. The expression reduces to a very compact form:

$$\boxed{\dot{m}(t) \;\approx\; \frac{V}{R\,T}\,\frac{dp}{dt}}$$

Numerical evaluation of $\dfrac{dp}{dt}$

With historic data points $\{\,p_i,\,t_i\,\}$ you can approximate the derivative by a finite-difference scheme, e.g.:

- **Forward difference** (simple, first-order)

$$\left.\frac{dp}{dt}\right|_{t_i} \;\approx\; \frac{p_{i+1} - p_i}{t_{i+1} - t_i}$$

- **Central difference** (second-order, less noisy)

$$\left.\frac{dp}{dt}\right|_{t_i} \approx \frac{p_{i+1} - p_{i-1}}{t_{i+1} - t_{i-1}}$$

Insert the resulting $\frac{dp}{dt}$ values into the constant-temperature formula above to obtain the instantaneous mass-flow rate $\dot{m}(t)$ at each sampled instant.

All symbols

- $p(t)$ – absolute pressure in the tank (Pa)
- $T(t)$ – absolute temperature in the tank (K)
- V – tank geometric volume (m³) – constant
- R – specific gas constant for dry air $\approx 287\ \mathrm{J\,kg^{-1}K^{-1}}$
- $\dot{m}(t)$ – instantaneous mass flow into the tank (kg s^{-1})